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(1) Research Program/Project/Study Title: Stock Differentiation of Genus <i>Upeneus</i> using Geometric Morphometric Analysis in Capiz, Philippines	
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Component Leader:	
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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
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☐ Applied	☐ Tech. Promotion/Commercialization								
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(9A) Abstract

The study of trait variation and its covariance with other variables is known as morphometrics (Rohlf and Slice, 1990). Nowadays, biological classification (Minton et al. 2008, Perez 2011, Miller 2016), botany (Su et al. 2021), and medicine (Du and Lu 2006) all make significant use of geometric morphometric (GM) approaches.

(9B) Keywords Geometric Morphometric Analysis, Genus *Upeneus*, Mahalanobis Distance, Canonical Variance Analysis, Procrustes Coordinates, Landmarks

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

It has long been believed that the fundamental idea of biology is the comparison of the anatomical characteristics of different living things (Medrano&Jumawan, 2016). The understanding of life and the classification of the organisms were found to be dependent on the morphological observations and description characteristics (Adams et al., 2004). The taxonomic identification of organisms and the scientific recording of the vast array on Earth confront significant challenges (Jaranilla, 2010). Due to these problems, scientists looked into other approaches for determining and examining the morphological attributes of each of the organisms (Adams et al., 2004). This gives birth to computer-based automated species recognition, which could assist taxonomy in overcoming this issue (Jaranilla, 2010). Morphometric measurement is one of the established and economically viable techniques for determining morphological distinctions among computerized species recognition systems (Juwan, 2014). This approach was created to be a valuable tool for quickly identifying species, particularly those in typical research (Eviota et al., 2016). Geometric morphometrics (GM) in biology has become a quickly developing discipline (Jensen, 1990; Bookstein, 1996; Rohlf, 1998). It is also considered the most suitable method for quantitatively comparing two-dimensional biological shapes (Cabuga et al., 2019). GM can either be a landmark-based analysis (using a set of landmarks to describe the object or specimen) or an outline-based analysis (using the margin of the specimen) (Richtsmeier et al., 2002). Historically, collections of quantitative variables, including length, width, and height, were subjected to multivariate statistical analysis in morphometrics (Adams, 2004). Significantly, this modern application analyzes the relative positions of anatomical landmarks and sets of points used to approximate curves and surfaces to quantify shape and size (Jensen, 2003). Rohlf and Adams et al effectively used GM analysis in their investigations to distinguish structural variation in Tuna. Among other functions, it is utilized to study species such as bamboo (Singh et al., 2013), snakehead fish (Requieron et al., 2012), soybeans (Meyer et al., 2006), and white stem borer (Albutra et al., 2012).

Species identification and population discrimination are critical in managing fisheries resources (Karakulak et al., 2016) and biodiversity conservation (Ibanez et al., 2007). It emerged that the morphological observations and descriptions of the organisms served as the foundation for both the taxonomy of the organisms and the understanding of life (Adams et al, 2004). Sheets (2001)

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states that morphometrics can be employed in quantitative studies, such as biological shape, shape disparity, and shape covariation alongside other biotic or abiotic variables. Investigating biological features, including growth, mortality, fecundity, trophic linkages, parasite relationships, and historical and paleontological occurrences, frequently needs accurate identification of individual specimens for the population (Ibanez *et al.*, 2007). According to Polly *et al.* (2016), geometric morphometric analysis may differentiate between closely related fish species and show a relationship between form and variations in developmental, evolutionary, functional, and ecological aspects (Imtiaz *et al.*, 2018).

Goatfishes, or Syngnatharia: Mullidae, are a varied family of fish prevalent and closely associated with coral reef environments. The family, which Linnaeus first described in *Systema Naturae* (Linnaeus, 1758), now has 102 recognized species spread across 6 genera (Fricke *et al.*, 2022). The crucial framework to investigate the intricate genetic and morphological relationships among the numerous closely related species has been provided by several genus-level phylogenetic studies of goatfishes (Turan, 2006; Keskin & Can, 2009; Uiblein & Heemstra, 2010; Uiblein, 2011; and Uiblein & Gouws, 2015). Morphological characteristics such as fin ray counts, body form, and dentition have been the main basis for goatfish taxonomy (Uiblein & Heemstra, 2011; Uiblein & McGrouther, 2012; Uiblein & Gouws, 2014; and Bos, 2014). A pair of highly developed hyoid barbels - which they use to detect and resuspend otherwise inaccessible food sources within the benthic substrate (Gosline, 1985)- which are fleshy appendages capable of chemoreception and prey excavation, are the primary synapomorphy that links goatfishes (Gosline, 1985). Goatfish species of the genus *Upeneus* frequently show overlapping distribution patterns within the Indopacific. Improved knowledge of the inter- and intraspecific morphological variability among *Upeneus spp.* should facilitate species identification and the search for functional relationships between body structure and resource utilization.

According to Monotora *et al.* (2011), eleven species of Genus *Upeneus* can be found in Panay Island. These are *Upeneus guttatus* (Quoy & Gaimard, 1825), *Upeneus japonicus* (Houttuyn, 1782), *Upeneus moluccensis* (Bleeker, 1855), *Upeneus stenosis* (Uiblein & McGrouther, 2012), *Upeneus subvittatus* (Temminck & Schlegel, 1843), *Upeneus sulphureus* (Cuvier, 1829), *Upeneus sundaicus* (Bleeker, 1855), *Upeneus tragula* (Richardson, 1846), and *Upeneus vittatus* (Forsskal, 1775).

The general objective of this study is to determine the stock differentiation of Genus *Upeneus* using Geometric Morphometric Analysis in Capiz, Philippines. Specifically, it aims to:

1. Determine the different body shapes of Genus *Upeneus* using Geometric Morphometric Analysis in Canonical Variance Analysis (CVA);
2. Delineate the body shape of the species under Genus *Upeneus* using Principal Component Analysis (PCA);
3. Identify the specific body shape of Genus *Upeneus* from pre-determined landmark

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(11) Materials and Methods or Methodology

Study Site

The study site was located at the coastal barangays of Capiz - the Seafood Capital of the Philippines - located in the Northern part of Panay Island. The barangays stated were the province's major source of fish and other marine resources. Samples were collected from fishing gear such as hook and line, gill net, etc, in the area with each sampling location as shown in Tab. 1 and its corresponding coordinates in Google Maps.

Table 1. Sampling sites and their corresponding coordinates

Locations	Longitude	Latitude
Basiao	11°32'55"N	122°38'28"E
Barra	11°35'09"N	122°43'06"E
Punta Cogon	11°36'34"N	122°47'35"E
Baybay	11°36'23"N	122°44'38"E
Libas	11°35'49"N	122°43'41"E
Culasi	11°36'32"N	122°43'41"E
Talon	11°33'56"N	122°40'47"E

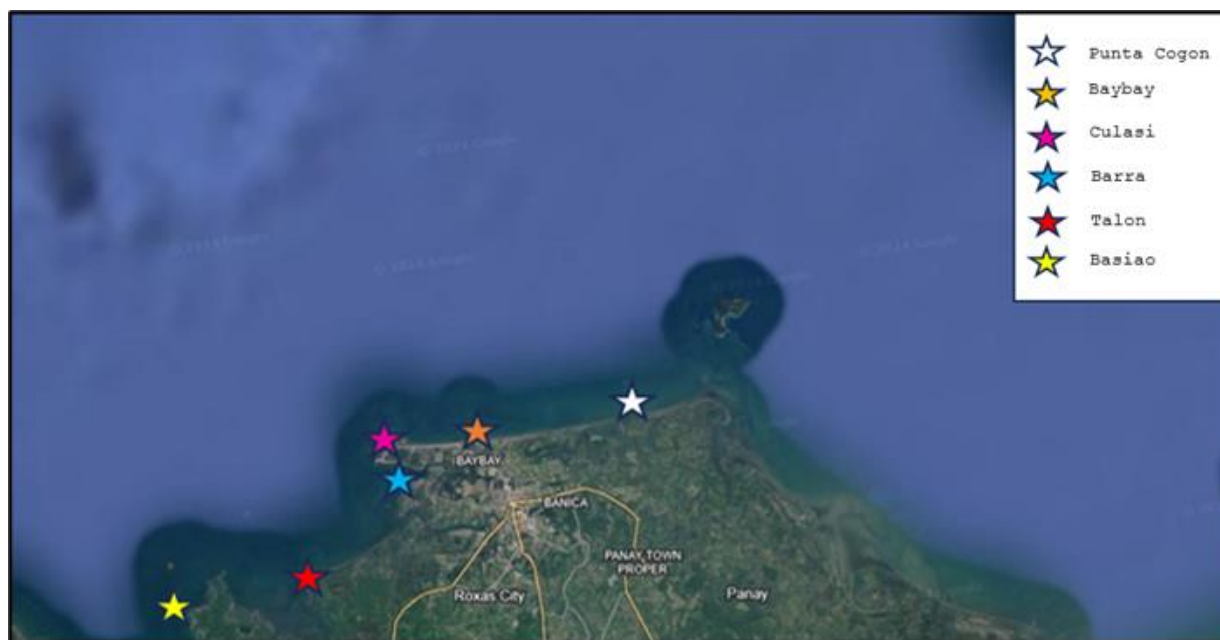


Figure 1. The sampling site of the study

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Sample Collection

A total of 400 samples of Genus *Upeneus* were collected from the different site locations from June 2023 to June 2024. Upon sampling collection, samples were transported to the Fisheries and Food Research Development Center Laboratory to be cleaned. Meristic characteristics were gathered and consolidated. Cleaned samples were then placed on the styroboard. The fins were spread properly, and 10% diluted formaldehyde was applied to the samples to keep the fins in place to mimic their posture underwater. The samples were photographed individually using a 108MP phone camera and compiled in a file to be processed in Thinplate Spline. Each of the samples was tagged and placed in sealed jars for preservation.

Species and Size Composition

Fish samples that were subjected to analysis are the species under the Genus *Upeneus*. According to Motomura et al. (2017), nine species belonging to Genus *Upeneus* are recorded in Panay Island. (1) ***Upeneus guttatus* (two-toned goatfish)** with a maximum length of 16 cm. It is described as having four reddish bars in the caudal fins. Color white or rose below the lateral line, covered by red pigmentation above the lateral line, which may also reach down ventrally and to the head, sometimes forming red patches or blotches; (2) ***Upeneus japonicus* (Japanese goatfish)** with the standard length of 16.2 cm. It is characterized by having an upper caudal-fin lobe in fresh fish with 3-4 red oblique bars, while lower caudal-fin lobe is almost entirely covered by a broad red stripe that becomes darker distally and is partially surrounded by a dark grey line and by a contrasting white fin margin and white lobe tip exteriorly; there are only weak indications of upper caudal fin lobe bars in preserved specimens with a dark pigmented zone remaining distally at the lower lobe; barbels are yellow; without lateral body stripes; head and body white or silvery ventrally and laterally, sometimes covered with red blotches, and uniformly red dorsally of lateral line; preserved fish are pale brown or brown (3) ***Upeneus moluccensis*** with the standard length of 22.5 cm. Observed of having red caudal fin bars on the upper lobe (faintly retained when preserved), none on the lower lobe but with a broad red band covering the entire lobe apart from the distal, inner margin, the latter somewhat darker (most of which are lost in preserved fish); one mid-lateral body stripe yellow or gold from eye to upper caudal-fin base (not or faintly retained in preserved fish); dark first dorsal-fin tip (retained in preserved fish); white barbels; silvery-rose body, dorsally darkened above lateral stripe (pale brown, slightly darkened dorsally in preserved fish) (4) ***Upeneus stenosis* (Narrow-tailed Goatfish)** is described by Uiblein & McGrouther (2012) as having a standard length of 13.1 cm. It is characterized by a total number of oblique caudal-fin bars 7-9, the color of bars changing from pale brown proximally to dark brown towards rear caudal-fin margin; dark bars on upper caudal-fin lobe 4-5 (including one bar close to the rear end of the lateral line), slightly increasing in width distally; lower caudal-fin lobe with 3-4 bars, of similar width as white interspaces between bars; first dorsal-fin tip black; caudal-fin bars and black dorsal-fin tip retained on preserved fish; barbels white; lateral body stripes absent; head and body silvery laterally and reddish dorsally, belly and ventral side of peduncle white; preserved fish dorsally or entirely pale brown or brown, with ventral part of body and head covered by the white or silvery pigment (5) ***Upeneus subvittatus* (Deep-water Goatfish)** with the standard length of

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24.1 cm (Golani, 2001) with plain, with silvery sides, a barred caudal fin and rounded snout, somewhat reminiscent of the silver oceanic or pelagic pufferfishes *Lagocephalus* that occur in the same area and represents a possible case of mimicry (Kuitert & Tono-zuka, 2001) (6) ***Upeneus sulphureus* (Sulphur Goatfish)** with the standard length of 23 cm (Sommer *et al.*, 1996). No bars on caudal-fin lobes; yellowish lower lobe base, grey on upper lobe (some of the caudal-fin lobe pigmentation retained when preserved, including inner margin of fin lobes); 2 yellow body stripes, 1 mid-lateral body from operculum to caudal-fin base (both faintly retained in preserved fish); white barbels; body dorsally pale brown and ventrally white (pale brown with some reddish or darker brown area at mid-body in preserved fish)(Kitchell *et al.*, 1997)(7) ***Upeneus sundaicus* (Ochre-banded goatfish)** with the standard length of 22 cm (Kumaran & Randall, 1984) characterized by 5-6 weak, red or grey bars on upper caudal fin lobe (not retained on preserved fish); no bars on lower lobe, fully to partly covered with a reddish or greyish band (lost in preserved fish); a pale brown mid-lateral body stripe from behind eye to caudal-fin base (sometimes retained in preserved fish); no dark dorsal-fin tip; yellow barbels when fresh; body reddish or dark grey dorsally, whitish-rose ventrally; head with red or grey pigmentation at snout and above eyes (body may darken dorsally in preserved fish) (8) ***Upeneus tragula* (Freckled Goatfish)** with the standard length of 30 cm (Karuppasamy *et al.*, 2019) 7–19 oblique bars on caudal fin, 3-9 bars on upper lobe, 4-10 on lower lobe; first dorsal fin with a large blotch around tip; one red, brown or black mid-lateral body stripe from tip of snout through eye to caudal base; body and head ground colour white or beige, slightly darker above the lateral line, with irregular red, brown, or black spots and/or blotches; yellow barbels but maybe pale brown or orange in fresh fish; most of the body and fin pigmentation is retained on preserved fish (Uiblein & Gouws, 2014) and (9) ***Upeneus vittatus* (Yellow-striped Goatfish)** with a standard length of 28 cm (Sommer *et al.*, 1996). 2 dorsolateral brown or pale brown stripes, with the lower one distinct and well-separated from pale body surface, extending from operculum to behind second dorsal fin, the upper one indistinct or hidden and much shorter, beginning below first dorsal-fin origin and bordered dorsally by a horizontal series of pale spots (lateral body stripes not retained on preserved fish); barbels white; white to silvery body. Dark reddish-brown dorsally, white belly, faint yellowish patches along pelvic and anal-fin bases (body pale brown in preserved fish, slightly darker above the lateral line)(Uiblein & Heemstra, 2010).

Landmark-based Geometric Morphometrics

A total of twenty-three landmark points were used in the shape analysis of the Genus *Upeneus* collected from the different sampling sites in Capiz, Philippines. Fig. 2 shows the location of the landmark points used in this study. Tab. 2 depicted the pre-determined landmark points and their respective description.

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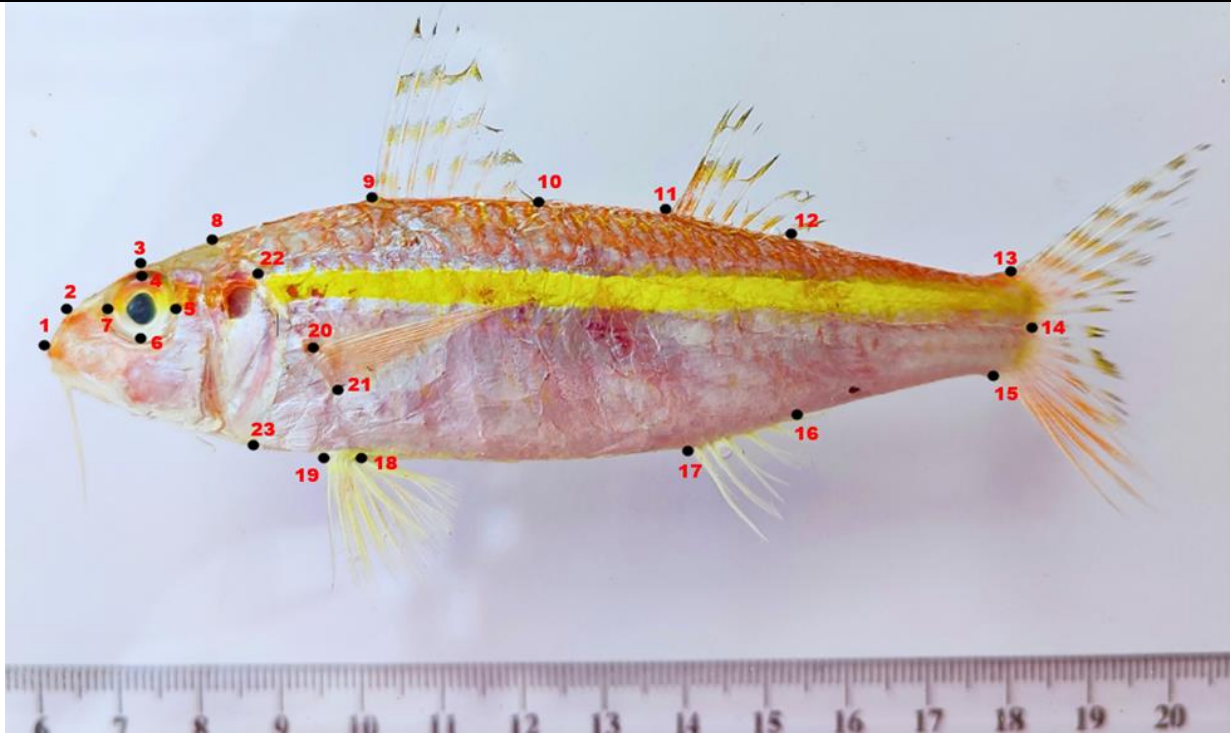


Figure 2. Landmark locations on the body of the actual sample (*U. moluccensis*) used to analyzed in this study.

Table 2. The landmarks and their corresponding description

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Landmark number	Landmark Description
1	Anterior tip of the snout
2	The nares
3	Dorsal Ventral side of the head above the eye
4	Dorsal side of the eye
5	End of the eye towards the tail
6	Ventral size of the eye
7	End of the eye towards the mouth
8	End of the cranium
9	Origin of the first dorsal fin
10	End of the first dorsal fin
11	Origin of the second dorsal fin
12	End of the second dorsal fin
13	Insertion of the upper lobe of caudal fin
14	Caudal peduncle
15	Insertion of the lower lobe of caudal fin
16	End of the anal fin
17	Origin of the anal fin
18	End of the pelvic fin
19	Origin of the pelvic fin
20	Origin of the pectoral fin
21	End of the pectoral fin
22	Insertion of the operculum
23	End of the lower lip

Data Analysis

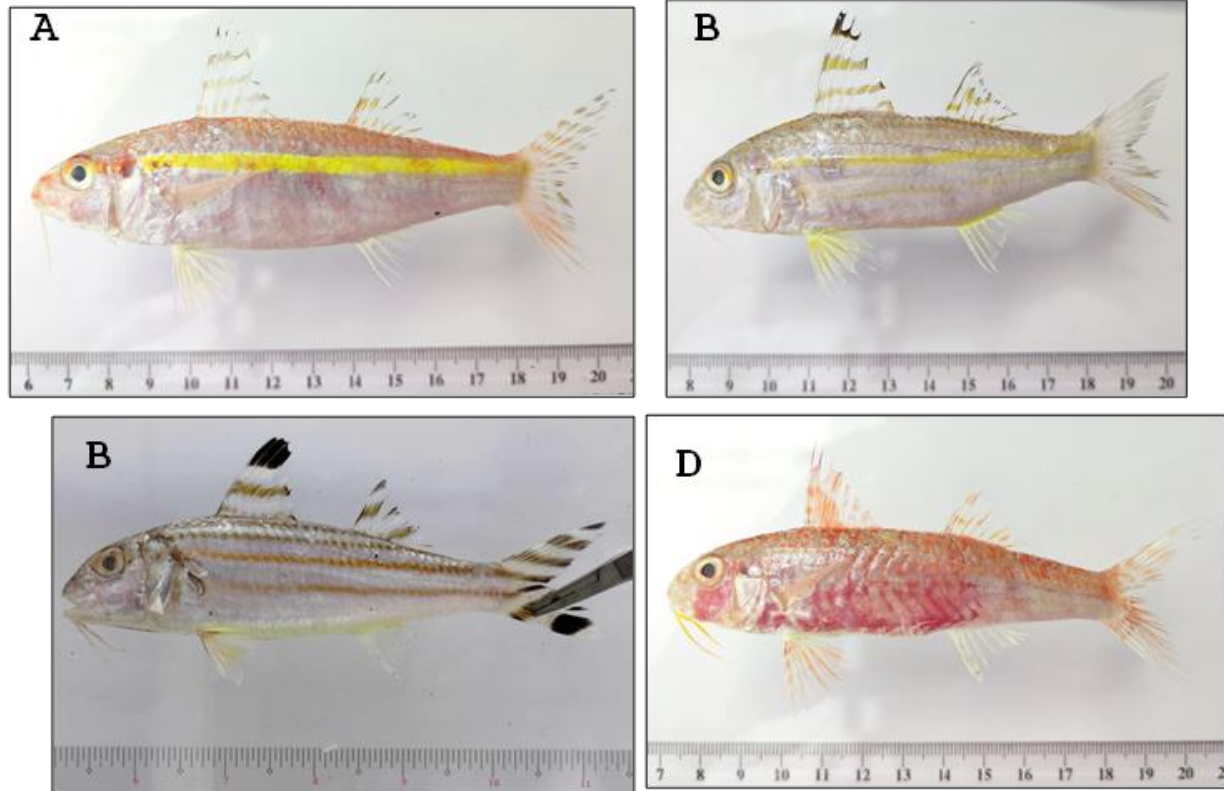
A total of 400 samples were collected from different sampling sites in Capiz, Philippines. Each of the samples were photographed. Photos obtained were compiled and subjected to data processing. Compiled photos were saved as a TPS file using a thinplate spline application (tps) application which controls and organizes the display, functioning, and storage of TPS files by applying the random scrambling scenarios to minimize bias in digitizing landmark locations, restoring the original order for repeated statistical analysis, creating empty files and adding keywords, etc. TPSUtil was used to convert the photos into TPS files. The files were subjected to digitalization and landmarks through the tpsDig application. TPSDig is a Windows (rough 11) program for digitizing landmarks and outlines for geometric morphometric analyses. Includes simple image enhancement operations, scale factors, image brightness profile, and support for AVI and MOV files. Output is in the TPS file format used by the "tps series" of programs. After digitalization, the files were then subjected to data analysis through MorphoJ. This application provides a user-friendly platform for the most important types of analyses in geometric morphometrics. The program is written for 2D and 3D data, with inevitable differences in the graphical output possibilities.

Common Species of Genus *Upeneus*

A total of 400 samples was used in this study collected from the different sampling sites of Capiz, Philippines from June 2023 to June 2024, 25% of which were identified as *Upeneus*

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sulphureus (Fig. 3B)- the most common species collected -, 25% of which is *Upeneus japonicus* (Fig. 3D), 25% of which is composed of *Upeneus moluccensis* (Fig. 3A), and 25% of the samples belong to *Upeneus vittatus* (Fig. 3B).



Variations among Species of Genus *Upeneus*

Canonical Variance Analysis (CVA) is a method used to find the shape features that best distinguish among multiple groups of specimens. The result of CVA (Fig. 4) depicts a correlation in terms of the body shape among the species of Genus *Upeneus*. Svanback (2016) suggested that GMA can differentiate the body structures of organisms. It also helps to discriminate the body structures and features of the fish samples (Cabuga, 2017). Based on the results, the samples belonging to *U. moluccensis* and *U. vittatus* are closely related with the Procrustes distance of 0.0516 with the P-value: <.0001. The Procrustes distance of *U. moluccensis* and *U. sulphureus* is 0.0545 with the P-value: <.0001. The result of *U. vittatus* and *U. sulphureus* shows the Procrustes distance of 0.0553 with the P-value: <.0001. The Procrustes distance between *U. sulphureus* and *U. japonicus* is 0.0724 with the P-value: <.0001. Between the Procrustes distance of *U. vittatus* and *U. japonicus* is 0.0730 with the P-value: <.0001, and lastly, the Procrustes distance of *U. moluccensis* and *U. japonicus* is 0.0933 with the P-value: <.0001. The results revealed that the samples of *U. moluccensis* and samples belonging to *U. vittatus* are closely related regarding body shape structure. It also determined that the *U.*

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japonicus and *U. moluccensis* show significant differences in body shape features. Notice that in Fig. 4, there is no overlapping of the samples, which indicates that the samples are of different species. The permutation test also supported these claims, which showed all samples have a P-value of $<.0001$. Regarding Mahalanobis distance (Tab. 3), it revealed that *U. moluccensis* and *U. japonicus* significantly differ from the MD of 178.2897. In contrast, the *U. vittatus* and *U. moluccensis* show low MD values of 92.2386, which means they are closely related to each other regarding body shape structures.

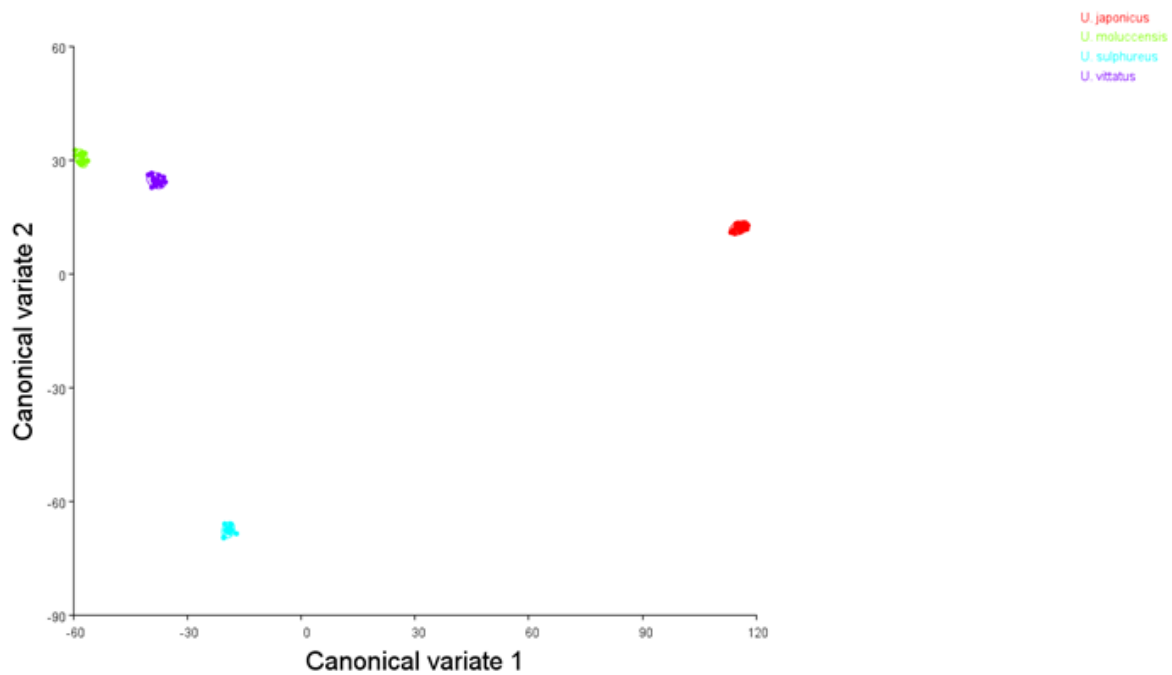


Figure 4. Plot of Canonical Variance Analysis of the 4 different sample species used in this study

Table 3. Mahalanobis distances between species of Genus *Upeneus*

	<i>U. japonicus</i>	<i>U. moluccensis</i>	<i>U. sulphureus</i>
<i>U. moluccensis</i>	178.2897		
<i>U. sulphureus</i>	156.6553	112.5564	
<i>U. vittatus</i>	163.3120	92.2386	106.7258

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Identifying the Shape Variation on the Species of Genus *Upeneus*

The variation of the sample species was identified through principal component analysis. PCA is a tool to understand overall patterns of shape variation and produces mathematically uncorrelated shape variables for subsequent analyses (Bookstein, 1997; Dryden and Mardia, 1998). It was utilized in this study to be applied to the complete dataset of Procrustes coordinates 42 principal component variations among samples belonging to Genus *Upeneus*. PC 1 and PC 2 displayed 81.994% cumulative values in Tab. 4.

Table 4. Principal Components, Eigenvalues, and Cumulative Percentages

PC's	Eigenvalues	% Variance	Cumulative %
PC 1	0.00115267	63.394	63.394
PC 2	0.00033820	18.600	81.994

Wireframe result of PC1 (Fig.5) showed that landmark 1 (anterior tip of the snout), landmark 2 (the nares), and landmark 8 (end of the cranium), located in the head region of the samples, were observed to contract towards the interior part of the body except to the landmark 1 that expands downward. These changes were observed to occur because of the feeding behavior of the species. The samples are demersal fish that live in sandy mud and muddy environments. The correlation between ecology and morphology in fish has long been recognized. Limited studies have utilized multivariate morphometric approaches to examine ecomorphological outlines in multi-species fish populations (Douglas ME, Matthews WJ. 1992). Landmark 13 (upper lobe of the caudal peduncle), landmark 14 (caudal peduncle), landmark 15 (lower lobe of the caudal peduncle), and Landmark 23 (end of the lower lip) were located at the posterior side of the fish, which responsible for the locomotion of the species. The variation between these landmarks was probably triggered by how they swim and escape from the predators.

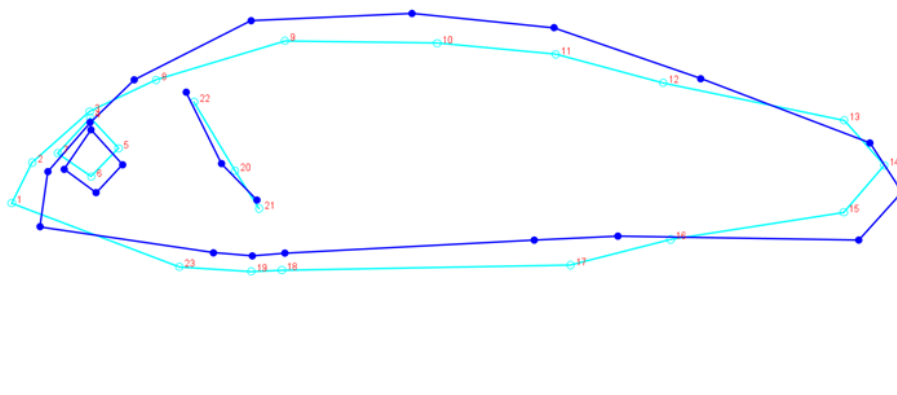
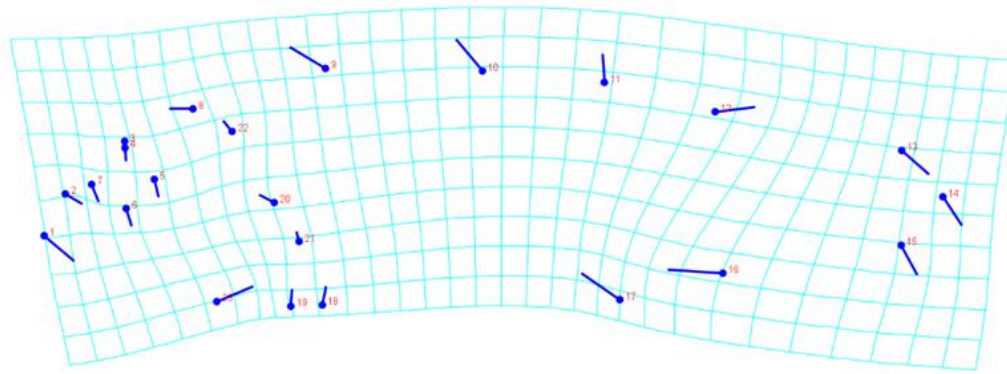


Figure 5. Wireframe figure of Principal Component 1

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PC1

Figure 6. The transformation grid of PC 1 depicts the shrinking and expansion of the landmark points

(13) Conclusions and Recommendations

Conclusions

This study was conducted to determine and differentiate the stock of Genus *Upeneus* from Capiz, Philippines. A total of 400 samples were collected from Brgy. Basiao, Brgy. Talon, Brgy. Barra, Brgy. Libas, Brgy. Culasi, Brgy. Baybay, and Brgy. Punta Cogon from June 2023 to June 2024. Four out of nine species of Genus *Upeneus* were collected and subjected to meristic analysis and geometric morphometric analysis. Upon sampling, *U. sulphureus* is the most abundant species. These species are always present during sampling, while *U. vittatus* is the last to collect.

The result of Canonical Variate Analysis revealed that there is a variation between species of Genus *Upeneus*. There is no overlapping of the populations of the samples, indicating that the samples are of different species. The CVA also revealed that *U. moluccensis* and *U. vittatus* are closely related, while *U. japonicus* displayed significant variation among the samples of Genus *Upeneus* with a Permutation value of $<.0001$. These claims were supported by Mahalanobis distance and Procrustes distances of the samples.

For the specific variation of the body shape of the samples, landmark 1 (anterior tip of the snout), landmark 2 (the nares), landmark 8 (end of the cranium), and landmark 23 (lower tip of the snout) located in the head region of the samples, were observed to contract towards the interior part of the body except to the landmark 1 that expands downward. These changes were observed to occur because of the feeding behavior of the species. The samples are demersal

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fish that live in sandy mud and muddy environments. Landmark 13 (upper lobe of the caudal peduncle), landmark 14 (caudal peduncle), landmark 15 (lower lobe of the caudal peduncle), and Landmark 23 (end of the lower lip) were located at the posterior side of the fish, which responsible for the locomotion of the species. The variation between these landmarks was probably triggered by how they swim and escape from the predators. Moreover, morphological distinctions amongst species can be plotted in a morpho-space where the organisms' distribution and occupied space support functional information on the structural complexity of the community (McClain & Johnson Rex, 2004; Clabaut *et al.*, 2007).

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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